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RESEARCH ARTICLE

Accurate Multiclass NLOS Channels Identification in UWB Indoor Positioning System-Based Deep Neural Network

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ABSTRACT The accurate distinction between line-of-sight (LOS) and non-line-of-sight (NLOS) propagation channels is paramount for precise distance measurement within ultra-wideband (UWB) indoor localization systems. In complex and dynamic environments, such as those encountered in the indoor positioning of autonomous mobile robots or vehicles, UWB signal propagation is particularly susceptible to NLOS conditions. However, much of the existing literature focuses on binary LOS/NLOS classifications, often overlooking the complexities of real-world environments such as hard-NLOS and multipath conditions. Additionally, a dynamic adaptation model for diverse indoor environments is lacking. This omission impedes the accuracy of UWB localization applications, such as autonomous robotics, where precision is vital in complex indoor settings. This article presents a fast hybrid-lightweight deep neural network model for LOS and NLOS conditions identification named indoor NLOS/LOS detection deep neural network. The proposed model ensures high accuracy and stability of LOS and NLOS identification with low processing time. More importantly, all of this comes with no loss, which is unique in the field of study. A robust verification has been guaranteed through experimental validation of four established databases. Moreover, the proposed framework performance is benchmarked against the recent LOS and NLOS state-of-the-art identification methodologies. The experimental findings underpin the efficacy and robustness of the proposed model in delivering an accuracy of 99.9% within a processing time of one second, which is the ideal recognition outcome validated across multiple databases.

INDEX TERMS Line-of-sight (LOS), non-line-of-sight (NLOS), ultra-wideband (UWB), channel impulse response (CIR), deep neural network (DNN), indoor positioning system (IPs).

I. INTRODUCTION

Indoor positioning, sometimes referred to as indoor localization, is a methodology that enables the creation of an exact position regarding the person or object inside an enclosed architectural space. Indoor positioning refers to a process of using several technological solutions that aim to allow a person to be precisely located inside buildings or other closed spaces using wireless communications, acoustic signal processing, infrared detection, and computer vision-based approaches. So much information has the potential to be applied in several fields of use. Some examples of applications that require accurate location include the asset tracking of some valuable assets within the Internet of Things (IoT), such as tracking personal devices. It includes healthcare systems for fall detection, enabling rapid assistance in case of falls among elderly or disabled individuals [1], automobile accident detection [2], interactive and technologically advanced museums [3], real-time surveillance of vehicle parking lots and autonomous payment [4], estimation of available parking spots using smart parking systems [5], monitoring of firefighters working under dangerous conditions

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such as dust, smoke, and fires [6], management of intelligent and location-based resources [7], access control based on the user's location, granting users access to resources according to their location [8], navigation within shopping centers [9], inventory and asset tracking in warehouses [10], [11], and identification of interior items, such as doors [12].

The majority of prevalent positioning techniques necessitate or presume a direct LOS path between the transmitter and receiver. Conversely, NLOS propagation, characterized by the obstruction of the direct path by an object, occurs through reflection, diffraction, or scattering. This phenomenon decouples the time of flight from the actual distance between the transmitter and receiver, thereby degrading the accuracy of time-based positioning methodologies. Similarly, NLOS paths can lead to erroneous estimation of the geometry between transmitter and receiver in angle-based positioning techniques. Consequently, prior to utilizing signal measurements in positioning algorithms, the classification of signals as either LOS or NLOS is imperative, particularly for applications where high accuracy is paramount.

The exploitation of this classification information can substantially enhance positioning accuracy and integrity. This can be achieved through various strategies, such as the selection of optimal positioning signals from a set of multiple signals by eliminating NLOS signals, alerting the positioning system when positioning would rely solely on NLOS signals, or dynamically switching between available positioning methods based on the prevailing LOS/NLOS conditions.

This article presents a method to identify UWB indoor LOS/NLOS multiclass conditions using a deep neural network. The proposed indoor NLOS/LOS detection deep neural network (INDDNN) model aims to achieve optimal accuracy in detecting NLOS conditions while minimizing complexity in both temporal and structural aspects. Additionally, it exhibits robust generalization capabilities across diverse datasets and environmental settings.

II. RELATED WORK

The infusion of a good number of deep learning and machine learning techniques has been performed in the field of drone communication localization, and navigation. Such methods are put to work to allow for greater accuracy and reliability in the classification and identification of signals under different environments. Researchers applied convolutional neural networks, recurrent neural networks, and transfer learning, among other original solutions, to the development of such challenging problems as non-line-of-sight conditions, signal interference, and environmental variability. While all these methods are highly successful, several problems persist, most of which have to do with computational complexity, application in real-time, and generalization within several environments. In this section, some major works done on this topic will be described, covering the methodology involved, the success achieved, and the limitations.

In 2023, Deng et al. proposed a parallel deep-learning model with multiple inputs to process data simultaneously. Their approach involves converting one-dimensional channel impulse response signals into two-dimensional images using Gramian Angular Field (GAF) technique. After that, a CNN extracts temporal features, while a residual network extracts visual features from these GAF-encoded images. It gains identification accuracy that varies across channels from a maximum of 96.05% to a minimum of 89.5%. However, the design incurs high computational resources due to reliance on one-dimensional signal and two-dimensional image inputs, which hinders real-time applications or deployment on less powerful hardware. This can further reduce the generalizability of the model in new scenarios, let alone in other UWB systems; it may need to be retrained or fine-tuned [13]. Sun et al., in 2022, present a method to classify three types of channel states within a single indoor environment. They use Stockwell transform (ST) and CNN to get features from both the time and frequency domains of UWB signal time-frequency maps. Then, they use transfer learning to make these features even better. A study utilizing open-source datasets and transfer learning with ST-CNN reports an F2 score of 97% for multi-class identification. However, this system's reliance on fine-tuning a small dataset based on UWB signal quality (time-frequency image) without considering other environmental factors in the training data may limit its performance in diverse indoor scenarios, even with transfer learning. Additionally, the use of a pre-existing model in transfer learning might constrain the understanding and control over representation learning for the new task [14]. In 2022, Wang et al. used one-dimensional wavelet packet analysis (ODWPA) to convert channel impulse responses into colored coefficient images (energy images) for classification by CNNs. The method achieved identification accuracy between 78.5% and 100% in three different scenarios with multiclass. However, its success depends on ODWPA's effective feature selection and transformation, which may not capture all relevant signal details. Additionally, the combination of ODWPA and CNNs adds computational complexity, which could limit real-time applications. Moreover, it may lose performance in scenarios that are not represented in the training data since its dependency is on characteristics specific to some indoor scenarios [15]. The authors of [16] presented a deep learning model that combined CNN with long short-term memory (LSTM) and fully connected layers for the classification of line-of-sight and non-line-of-sight scenarios using thirteen input variables. The input variables for the model encompass the threshold for identifying the initial path, amplitude values of the three paths adjacent to the first path, noise level standard deviation, channel impulse response maximum growth, accumulated preamble symbol count, estimated locations of the first and peak paths, the amplitude of the peak path, the difference between received signal strength and first path power, the difference between first path and peak path locations, and skewness surrounding the first path. While the accuracy and F1 scores reported were rather high, 95%–98%, the model substantially overlooked the effects of multipath and hard

non-line-of-sight [17]. This model presents a complex structure because the combination of CNN and LSTM will cause computation and time overheads. This will limit the applications that require real-time processing or where computational resources are limited. Generalization to very diverse indoor environments will also not be possible to a great extent since it has been validated only in a specific indoor environment by overlooking multipath and hard-NLOS effects. In [18] 2023, the authors proposed a Bayesian-optimization-based fuzzy decision tree approach. In this methodology, eight-channel characteristics were derived from the CIR. The features include total energy, maximum amplitude, normalised strongest path energy, rising time, signal-to-noise ratio, mean excess delay, root-mean-square delay spread, and kurtosis. Although the reported NLOS identification accuracy was high in two experiments for binary classification, the large discrepancy between the results (89% and 98%) may indicate that the system is not very stable and does not generalize to all scenarios. Moreover, the performance of the BO-FDT seems to be sensitive to the size of the training data set. More importantly, Bayesian optimization in itself is based on the proper choice of prior and likelihood functions. Therefore, careful consideration has to be placed on the choice of these parameters to best adapt to the characteristics of the data and model at hand. Yang et al. In 2023 proposed an approach for improved LOS/NLOS identification accuracy, utilizing new parameters extracted from channel impulse response data. The three new CIR characteristic parameters were: the number of false crests, the first path error, and the first path distance error, which were obtained from the waveform and energy of critical nodes. Their two-step classification process combines dynamic threshold comparison and a fuzzy credibility-based support vector machine. While experimental results show a 93% accuracy improvement, the system's reliance on precise Characteristic Parameters (CCP) thresholds may limit its robustness to environmental variations. Further testing in diverse settings is needed to assess its broader applicability [19]. The authors of [20] suggested a new deep-learning architecture called ININN. This hybrid model integrates a CNN for spatial feature extraction, a gated recurrent unit (GRU) for temporal feature extraction, and squeeze-and-excitation (SE) blocks for channel-wise feature enhancement. They utilise energy detection to ascertain the initial path of the CIR signal as a time of arrival. While the system achieves an accuracy of 88.45% on an open-source dataset and demonstrates a variable accuracy range (87.5% to 99.9%) under different energy per bit to noise power spectral density ratio (E_b/N_0) conditions, its performance is susceptible to environmental factors and signal quality. Notably, including CNN, GRU, and SE components increases computational complexity, potentially limiting real-time applicability on resource-constrained devices. Further investigation is warranted to address the observed performance variability and optimize the model for real-world deployment. Wei et al. in 2022 introduced a multiple-input learning neural network that processes both raw CIR signals and time-frequency diagram of CIR (TFDOCIR) images generated using continuous wavelet transform (CWT). This system demonstrates an identification accuracy of 91.74%. Nevertheless, the model's complexity, which integrates GRU and CNN layers, necessitates considerable computational resources and extended training times, posing challenges for deployment in real-time or resource-constrained environments. Furthermore, the accuracy of NLOS identification is highly dependent on the precise generation of TFDOCIR images from CIR signals; any errors in this preprocessing stage can significantly degrade the overall system performance [21]. In 2022, Che et al. introduced a feature-based method to enhance the accuracy of NLOS detection, which focused mainly on the problem caused due to the imbalanced datasets. In this system, gaussian distribution (GD) and generalised gaussian distribution (GGD) are used. The GD algorithm is based on the mean and variance of the features extracted, while the GGD uses kurtosis, which models the distribution of data more appropriately. It recorded an accuracy of 98% in binary classification, especially in imbalanced datasets. However, they evaluated the system's performance using a small dataset in a specific environment with a range not exceeding seven meters, employing a dynamic threshold that adapts based on the training data. The system is still sensitive to variations in different real-world indoor settings. That is, generalizability and robustness need further tests on a larger and more diversified dataset [22]. In 2020, the authors of [23] put forward a method that combines morlet wavelet transformer (MWT) with CNN, called MWT-CNN, which transforms the UWB CIR data into time-frequency spectrums through MWT and then processes them with a CNN to extract the features for determining the NLOS conditions. The extracted features include energy, maximum rise time, mean excess delay, root-mean-square delay spread, and kurtosis. It also has an accuracy as high as 98.84% in scenarios quite close to the training environments. It has heavily degraded performance on the other testing scenes, reaching an accuracy of 76.65% in the industrial scene and 69.23% in the residential scene. Consequently, MWT-CNN requires a very specific stable environment. The limited generalization capability, along with the high computational complexity and large requirements on training data of this method, pose significant challenges for broader and real-world applications [23]. Wang et al. in 2020 proposed a semi-supervised support vector machine classifier (S3VM) that includes a self-training mechanism; an initial classifier is trained using a small set of labeled data extracting several waveform features. This classifier is then used to pseudo-label the unlabeled data iteratively added to the training set according to some measure of confidence. While the proposed system represents a 94% improvement over supervised learning methods, these incremental benefits may not justify additional complexity and computational resources for S3VM. Further, due to its vulnerability to biases and

insufficiencies in the initial labeled data, the model can generalize very little without retraining. Moreover, the performance is sensitive to parameter selection for instance, the weights between labeled data and pseudo-labeled data, and the scale factors in iterative training. The input variables for the model include the estimated distance, signal energy, maximum amplitude, rise time, mean excess delay, root mean square delay spread, and kurtosis [24]. The authors of [25] used machine learning to identify multiple-class NLOS conditions in ultra-wideband indoor localization systems. In the experiment, data were collected to evaluate the performance of three machine learning classifiers: SVM, random forest (RF) based on the ensemble learning method, and multilayer perceptron (MLP). The overall accuracy of these classifiers is significantly reduced when tested in an environment different from the one in which the training was performed. For example, the RF classifier accuracy drops from 91.9% to just 73.5% across environments. The same thing happens with SVM and MLP classifiers, suggesting generalization due to the lack of representation by diverse conditions within the training dataset. Performance degradation when testing in different environments strongly suggests overfitting to the training data. While learning the specific features, the classifiers obtain very specialized knowledge of the training environment and thus do not generalize well to new unseen ones. The experimental assessment of this study was predicated on three parameters: the observed distance, the amplitude of the CIR signal, and the amplitude of the first harmonics in the primary path signal [25]. Cui et al., in 2020 utilized CNN to binary classify various types of CIR figures, thereby converting the identification problem into one of image recognition. Similarly, [27] employs capsule networks (CapsNets) for the same task. The CIR is processed using a short-time Fourier transform (STFT) to generate an impulse response spectrum, which serves as input to the CNN, achieving an accuracy of 98.24%. In the CapsNets architecture, which comprises a convolutional layer, a primary capsule layer, and a channel capsule layer, the input data is formatted as a $6\times6\times1$ monochrome image. CapsNets utilize a dynamic routing-by-agreement mechanism to iteratively refine the coupling coefficients between capsules, resulting in an accuracy of 94.63%. Nevertheless, the performance of both models is significantly influenced by the quality and quantity of the training data and is sensitive to hyperparameter selection. This necessitates meticulous tuning and additional validation across diverse indoor environments, which can hinder generalization to new datasets. Furthermore, these models are more computationally intensive compared to traditional machine learning models, requiring the preprocessing and structuring of input data into image format [26]. The input parameters employed in these approaches include mean excess delay, the count of rays with amplitudes exceeding a threshold, the number of rays contributing 85% of the total energy for training and testing set generation, root-mean-square delay spread, observed distances, real distance, error, energy, maximum, rise time, and kurtosis. The authors of [28] suggest a system that integrates feature extraction from the CIR with machine learning algorithms, employing SVM optimized via a genetic algorithm (GA) for feature selection. SVM is utilized to classify signal propagation conditions as either LOS or NLOS. The GA is employed to identify the optimal combination of features that maximizes classification accuracy while minimizing computational complexity. The selected optimal feature set comprises root-mean-square delay spread, maximum amplitude, received signal energy, reported distance, and the time delay from pulse peak to start. The proposed system achieves an NLOS detection accuracy of 91.86% with the optimal feature combination. Y. Pei et al. in 2023 presented an FCN-Attention deep learning algorithm that integrates a fully convolution network (FCN) for feature extraction with a self-attention mechanism in a synergistic way for feature enhancement. The efficacy of this method was rigorously evaluated across three distinct datasets: the eWINE Project Dataset, the Local Dataset, and a Mixed Dataset, yielding classification accuracies of 88.24%, 100%, and 92.01%, respectively. A notable limitation of the FCN-Attention algorithm is its dependence on the quality and diversity of the training data, which is prominently demonstrated by the variability in identification accuracy across different experimental scenarios [29]. Table 1 presents the summary of related works that depend on deep learning techniques. Based on the comprehensive review of current state-of-the-art methodologies, it is evident that simultaneously addressing all the discussed challenges remains a formidable task. To the best of the authors' knowledge, no model has been developed that effectively enhances both accuracy and time efficiency while considering the necessary trade-offs. This article introduces a distinctive framework designed to achieve optimal accuracy, thereby making significant contributions to the field of rapid deep learning for multiclass LOS/NLOS condition identification.

This work critically reviews and evaluates cutting-edge research in the domain. A hybrid deep neural network with efficient pre-processing techniques is proposed, specifically developed for indoor positioning systems utilizing the UWB DW1000 device. The proposed INDDNN model's performance and measurement parameters were rigorously assessed against several state-of-the-art methods and validated using four open-source databases, in addition to an experimentally collected dataset.

III. PROBLE FORMALATION

The principal objective of the identification process in wireless communications is to discern the presence of line-of-sight (LOS) or non-line-of-sight (NLOS) conditions between a transmitter and receiver. This step is critical, as NLOS conditions significantly degrade the accuracy of distance. However, practical applications face significant challenges, notably the NLOS problem, which causes signal delays and biased distance estimates, degrading positioning accuracy.

NLOS conditions arise when the direct path between the signal transmitter and receiver is obstructed.

TABLE 1. Summary of related studies.

Reference No.	Year	Used Technique	Limitations
[13]	2023	A parallel model of CNN and residual network for deep learning.	Limited real-time applicability; problems in generalizability.
[14]	2022	ST-CNN with transfer learning	Limited performance in most scenarios, strong reliance on fine-tuning
[15]	2022	ODWPA with CNN	Computational complexity, environment-specific performance
[16]	2023	CNN-LSTM model	High computational overhead, limited generalization
[18]	2023	BO-FDT approach	System stability issues, training data size dependency
[19]	2023	Two-step classification with fuzzy SVM	Environmental robustness concerns
[20]	2022	ININN model with CNN-GRU-SE	High computational complexity, performance variability
[21]	2022	Multiple-input learning neural network	High computational resources required, preprocessing dependency
[22]	2022	Feature-based method with GD and GGD	Small dataset, environment-specific performance
[23]	2020	MWT-CNN method	Limited generalization, computational complexity
[24]	2020	S3VM classifier with self-training	Incremental benefits vs. complexity, generalization issues
[25]	2020	Machine learning classifiers (SVM, RF, MLP)	Generalization problem, overfitting
[26] [27]	2020	CNN and CapsNets	Sensitivity to training data quality, high computational needs
[28]	2019	SVM with GA for feature selection	Specific to optimal feature combination
[29]	2023	FCN-Attention algorithm	Dependent on training data quality

Furthermore, the existing literature focuses mainly on binary LOS/NLOS classification. This binary approach ignores the complexities of real-world environments, especially in challenging scenarios involving hard-NLOS and MP. Moreover, current approaches lack dynamic adaptation to diverse indoor environments, resulting in models with limited generalization. Therefore, accurate NLOS identification is crucial for precise indoor positioning, especially for autonomous navigation, search-and-rescue operations, and industrial tracking where sub-meter accuracy is essential. In a UWB localization network, two types of nodes exist, tags, whose states are initially unknown, and anchors, whose states are consistently known.

Fig. 1 also depicts four propagation scenarios. In the first scenario (LOS), there are no obstacles between the transceivers. The second scenario (MP) involves multiple reflections distorting the ranging signal. In the third scenario, a single obstacle (a wall) blocks the direct path signal between the anchor and Tag Two. The fourth scenario presents two obstacles obstructing the direct path signal between transceivers, resulting in increased signal travel time and causing direct path excess delay. Under NLOS conditions, when the ranging signal penetrates or reflects off walls, the change in the propagation medium distorts the signal waveform and reduces the power level of the first path received.

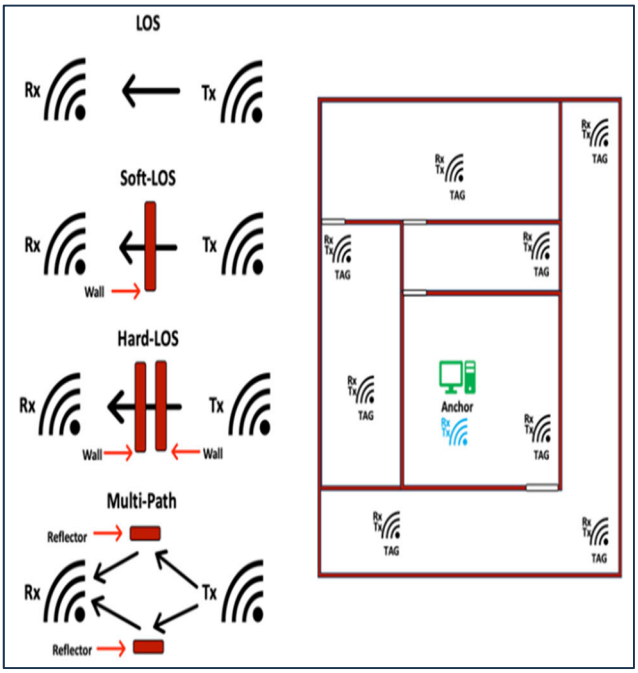


FIGURE 1. Indoor environments.

IV. SYSTEM METHODOLOGY

The INDDNN methodology necessitates the availability of channel impulse response (CIR) data for each ranging measurement that is to be classified as LOS or NLOS. The feasibility of the proposed system in different environments was investigated by relying on multiple features that were extracted from the DW 1000 to create an authentic

database that complies with IEEE 802.15.4a standards. Categorical labels were converted into numerical values and subjected to normalization and outlier removal procedures to enhance the robustness of the subsequent analysis. Thereafter, our INDDNN is implemented. Fig. 2 illustrates the system architecture of the proposed model, encompassing the data creation, preprocessing, the INDDNN model, and the evaluation/validation phases. A detailed description of this architecture follows in the subsequent subsections.

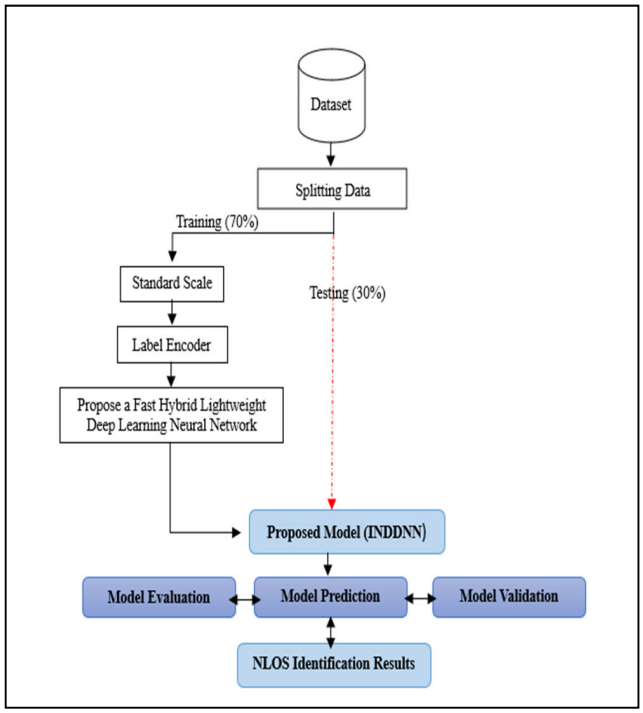


FIGURE 2. Diagram of the proposed INDDNN model architecture.

A. DATA COLLECTION

The experimental evaluations concentrated on UWB data measurement and used the Qorvo DW1000 module as our UWB hardware [30] for data creation. The Qorvo DW1000 can work well in high reflective RF conditions due to its resistance to rigorous multi-path conditions. This module is known to provide accurate and reliable UWB signal transmission and reception; hence, it was chosen for our research. Table 2 shows the hardware configurations used for the experimental evaluations. In the measurement phase, the database collection process created a basic network architecture to satisfy the objectives of this study. It was made up of two nodes for the measurements, each with a Qorvo DW 1000 UWB transceiver. This simple configuration of this network served as a base on which to develop the generation and collection of the database. One UWB transceiver will work as an anchor, and the other one as a target. The anchor connects to a computer for data logging over a serial USART port. In this way, UWB signal data during the experiments can be captured and recorded with accuracy. Software for two-way ranging was developed for production testing of Qorvo Decawave's evaluation kit EVK1000; this worked for both transceivers. This software is available to the public on the official website of Qorvo Decawave. Some necessary changes have been made in the library of the software to fit our experiment's objective, specifically for feature extraction. These adjustments enabled us to extract all the necessary features identified for our analysis. Following the extraction, the database, including the identified features for each measurement trial, was systematically logged and stored in a file (utilizing MATLAB R2022a). This methodical approach ensured that our dataset was well organized and comprehensive, capturing a wide array of signal characteristics essential for our study. The antenna height of the static UWB device was consistently kept at 2.00 m to reduce the effect of Fresnel zones on the measurement accuracy. This device directly records the measurement data on a connected PC. Ensuring meticulous attention to detail was crucial in ensuring the representativeness of our experimental results regarding the performance of UWB systems in various environments, as well as enhancing the reliability of the measurement outcomes.

TABLE 2. Specifications of the DW1000 module.

Module name	DW1000
Manufacturer	Qorvo
Channel	2
Frequency	4 GHz
Bandwidth	500 MHz
Data Rate	110 Kbps
Pulse Repetition Frequency	64 MHz
Preamble length	1024
Preamble cod	9
Reported precision	10 cm
Microcontroller type	STM32L476RG
Microcontroller Manufacturer	STMicroelectronics

B. MEASUREMENT ENVIRONMENT

The real measurement data were conducted in two residential environments encompassing multiple rooms that were selected for the measurement campaign to cover the variable influence of indoor environments on RF signal propagation. The data required for our experimental evaluations were gathered in four rooms of varying dimensions (5 m × 4 m, 10 m × 4 m, 20 m × 6 m, and 4 m × 2 m) within a building environment, a hall measuring 8 m × 48 m containing various items of furniture, and two corridors with widths of 3 m and 1.5 m, respectively. These measurement data were collected under different propagation channels across these diverse scenarios. For the NLOS experiments within our project, the setup required the creation of obstacles between the anchor and target transceivers to simulate NLOS conditions, necessitating the use of two different rooms. In particular, this work has focused on the analysis of soft and hard NLOS propagation channels, which helps in understanding how different obstacles may affect UWB signal transmission, as depicted in Fig. 1. Fig. 3 shows the real environment wherein data

has been collected under different propagation channels. The illustrations describe the scenario in which the signal emitted from a tag device is received by anchor nodes under several conditions in a situation where an UWB transceiver, Qorvo DW 1000, acts as both the tag device and the anchor nodes.

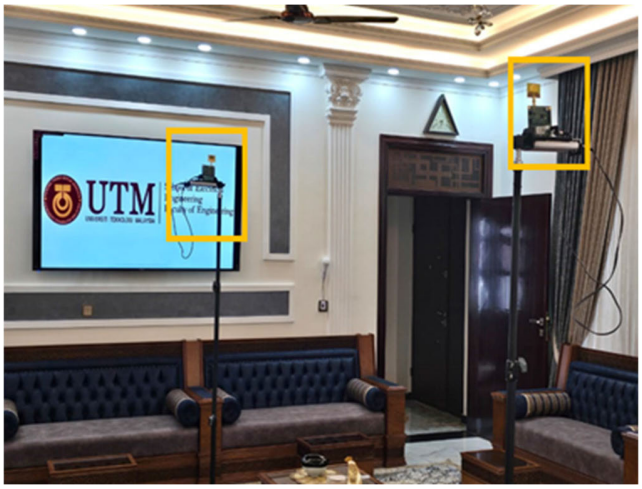


FIGURE 3. Ultra-wideband (UWB) localization system.

C. MEASUREMENT PROCEDURE

In the present study, great care was taken while designing the methodology for collecting data at both static and dynamic scenarios that would help in properly analyzing the signal behavior for a wide range of conditions. In the dynamic scenarios, the experimental setup was configured so that one transceiver was fixed and connected to a PC device for logging data while its counterpart was made mobile and given to a person performing random locomotion. Such a design is envisioned to simulate real-life situations in which there are frequent position and orientation changes of the target device. In the static setup, the configuration provided that both transceivers be in an upright stance, forming a perpendicular angle concerning the ground plane, and mounted on a 2m tall tripod with the Qorvo DW1000 antenna oriented vertically. Such a configuration would ensure the same orientation of both devices to provide an environment that was controlled for measurements of propagation of the UWB signal in scenarios with no movement. The cellular transceiver antenna was spinning intermittently during the collection period under dynamic conditions at random intervals. The induced variability in the orientation of the antenna was one way of evaluating the associated effect of device orientation on signal strength and measurement quality, broadly simulating a more practical scenario whereby the device would be oriented at random. The data collection process was set up to evaluate in detail the operational performance of the UWB system in the indoor environment involving various environmental conditions. Adding movement and different orientations to the dataset made it much more comprehensive, which led to deeply insightful discoveries about the system's adaptability and effectiveness in real-life situations. A total of 12 features were extracted from the real measurements recorded on the range path between transceivers. These measurements encompassed four distinct propagation conditions: LOS, MP, soft-NLOS, and hard-NLOS. The respective number of measurements for each condition, per feature, were as follows: LOS (48812), MP (36365), soft-NLOS (44104), and hard-NLOS (18454). Table 3 enumerates the configuration settings of the model for the UWB devices' DW1000 model.

TABLE 3. Settings of UWB models.

Scenario	Channels	Distances	Range Cover	No. Samples	CIR Features
Indoor residential, office, Industrial and outdoor	LOS	19	Up to 16m	48812	12
Indoor residential, office, and Industrial.	MP	18	Up to 13m	36365	12
	Soft-NLOS	20	Up to 15m	44104	12
	Hard-NLOS	9	Up to 10m	18454	12

D. DATA PROCESSING AND FEATURE EXTRACTION

In the data preprocessing phase following the measurement campaign, the class labels (LOS, MP soft-NLOS, and hard-NLOS) were manually assigned. For each experimental trial, observations corresponding to the four categories were systematically recorded and stored on a PC as individual files. Specifically, these files represented collections of data from identical class types, organized separately. This method facilitated the straightforward labeling of entire data sets as LOS, MP soft-NLOS, or hard-NLOS based on the scenario being evaluated. Post-collection, class labels (LOS, MP soft-NLOS, and hard-NLOS) were manually assigned. Categorical labels are then converted into numerical values for the learning algorithms. The data was then segregated and stored as individual files for each class. Feature scaling was performed using standardization, rescaling the data during preprocessing to a mean of zero and a standard deviation of one (unit variance). This normalization of input features typically results in faster training times and improved performance for many machine-learning models. In total, twelve features were extracted from the Qorvo DW 1000 UWB modules [31], following the configuration outlined in Table 3. These features encompassed all available parameters that could be collected from the Qorvo DW 1000. The features, derived from our publicly available IEEE Dataport datasets [32], include: raw timestamp, accumulated preamble symbols (PSC), received signal strength level (RSL), first path signal strength (FSL), noise threshold values, a numerical representation of the signal's leading edge, range correction values (Bias), timing difference between the tag and anchors, measured distance, clock offset, and the receiver's internal re-sampler time delay

TABLE 4. Ranging errors across various channels.

Actual Distance (m)	LOS		Soft-NLOS		Hard-NLOS		MP	
	Measured Distance (m)	Bias (m)	Measured Distance (m)	Bias (m)	Measured Distance (m)	Bias (m)	Measured Distance (m)	Bias (m)
1	1.03	0.03	1.41	0.41	..	..	..	..
2	2.00	0.00	2.37	0.37	..	..	2.33	0.33
3	3.03	0.03	3.34	0.34	4.17	1.17	3.47	0.47
4	4.07	0.07	4.35	0.35	5.22	1.22	4.46	0.46
5	5.07	0.07	5.34	0.34	6.02	1.02	5.31	0.31
6	6.06	0.06	6.34	0.34	7.13	1.13	6.55	0.55
7	7.05	0.05	7.34	0.34	8.02	1.02	7.58	0.58
8	8.03	0.03	8.32	0.32	9.02	1.02	8.27	0.27
9	9.08	0.08	9.29	0.29	10.23	1.23	9.61	0.61
10	10.02	0.02	10.36	0.36	11.19	1.19	10.32	0.32
11	11.02	0.02	11.29	0.29	12.21	1.21	11.53	0.53
12	12.05	0.05	12.34	0.34	13.21	1.21	12.39	0.39
13	13.05	0.05	13.25	0.25	14.20	1.20	13.23	0.23
14	14.04	0.04	14.41	0.41	15.22	1.22	14.40	0.40
15	15.06	0.06	15.27	0.27	16.21	1.21	15.38	0.38
16	16.09	0.09	16.34	0.34	17.21	1.21	16.40	0.40

(RSMPL). A feature selection process was conducted using the wrapper technique to identify the most relevant attributes for NLOS identification. Among the features considered, RSL, FSL, PSC, and measured distance were found to have a statistically significant impact, which were employed in the proposed system. It is pertinent to note that the extraction of these features was exclusively conducted using the Qorvo DW1000 chip, a piece of UWB hardware produced by Qorvo. The process was facilitated by using Qorvo software version 3.11 in conjunction with MATLAB R2022a. This methodological choice underscores the integration of specialized hardware and sophisticated software tools in the feature extraction process, ensuring the precision and relevance of the features derived for the study.

Table 4 provides a comprehensive analysis of the bias in measured distances across the 1-16 m range, offering an in-depth overview of the influence of various propagation channels on ranging and accuracy. This analysis yields valuable insights into how different visibility classes, such as LOS, MP, and both soft and hard NLOS, impact the overall positioning system performance.

V. PROPOSED DEEP LEARNING METHOD

A. PROPOSED INDDNN MODEL

Deep learning, a primary subset of machine learning, is characterized by its architecture, which employs multiple layers of non-linear information processing arranged hierarchically. This approach is particularly advantageous for analyzing complex datasets [33]. A large amount of data is required for a DNN to extract features and train them effectively to identify the patterns and create meaningful weights. A traditional DNN is developed with special hardware that has high specifications since the raw data used as input requires no pre-processing or feature extraction operations; hence, the training process involves a significant amount of time.

This paper presents a fast hybrid lightweight deep neural network for multiclass LOS/NLOS condition identification. The proposed INDDNN is a one-dimensional neural network adapted from the traditional two-dimensional neural network. This method, owing to its simplicity in computations and speed, can be executed in cost-effective and simple environments, making it useful for practical applications. This application will feed each input through repeated convolutional layers with kernels, pooling, and fully connected layers to classify the LOS/NLOS conditions with very high accuracy. The nine one-dimensional convolutional layers belonging to the INDDNN having 23 layers are tasked with the extraction of prominent features. Eight max-pooling layers to enhance the model's translational invariance, ensuring that minor spatial shifts in the input do not significantly alter the network's output and extract the most salient features from the preceding convolutional layer, effectively reducing the dimensionality of the representation while preserving the most critical information. Four dense layers, employing ReLU activation functions, function as feature collectors. ReLU's linearity makes it easier to collect stronger features, which increases the number of features collected. One flatten layer converts the multidimensional output from the preceding layers into a one-dimensional vector, ensuring that its

structure matches the dimensionality of the initial input data. The final Dense Layer with SoftMax Activation Function is used to categorize the features derived from the previous layers. The SoftMax function converts these features into values between 0 and 1, enabling their interpretation as probabilities. Inputs that are small or negative are transformed into low probabilities, while larger inputs are converted into higher probabilities, yet they consistently remain within the 0 to 1 range. The model is trained using the Adam optimizer with a learning rate of 10^{-3}, a batch size of 512, and is executed for 500 epochs. Table 5. illustrates the layers of the INDDNN model with more information. Additionally, Fig. 4 visually represents the model's architecture, consisting of 23 layers, including nine one-dimensional convolutional layers, eight max-pooling layers, five dense layers, and one flattening layer.

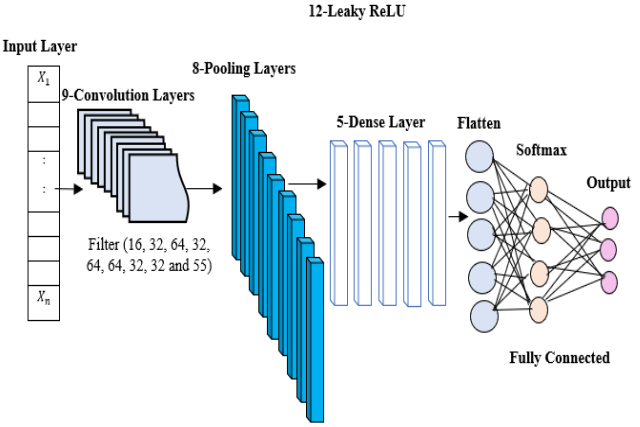


FIGURE 4. The proposed INDDNN model.

TABLE 5. Layers of the proposed INDDNN model.

NO.	Layer	Parameter	Output-shape
1	conv1d (Conv1D)	64	(None, 10, 16)
2	max_pooling1d (MaxPooling1D	0	(None, 10, 16)
3	conv1d_1 (Conv1D)	1568	(None, 10, 32)
4	max_pooling1d_1 (MaxPooling1	0	(None, 10, 32)
5	conv1d_2 (Conv1D)	6208	(None, 10, 64)
6	max_pooling1d_2 (MaxPooling1		
7	max_pooling1d_3 (MaxPooling1	0	(None, 10, 64)
8	conv1d_3 (Conv1D)	6176	(None, 10, 32)
9	max_pooling1d_4 (MaxPooling1	0	(None, 10, 32)
10	conv1d_4 (Conv1D)	6208	(None, 10, 64)
11	max_pooling1d_5 (MaxPooling1	0	(None, 10, 64)
12	conv1d_5 (Conv1D)	12352	(None, 10, 64)
13	max_pooling1d_6 (MaxPooling1	0	(None, 10, 64)
14	conv1d_6 (Conv1D)	6176	(None, 10, 32)
15	conv1d_7 (Conv1D)	3104	(None, 10, 32)
16	max_pooling1d_7 (MaxPooling1	0	(None, 10, 32)
17	dense (Dense)	4224	(None, 10, 128)
18	dense_1 (Dense)	8256	(None, 10, 64)
19	dense_2 (Dense)	4160	(None, 10, 64)
20	dense_3 (Dense)	1040	(None, 10, 16)
21	conv1d_8 (Conv1D)	2695	(None, 10, 55)
22	flatten (Flatten)	0	(None, 550)
23	dense_4 (Dense)	1653	(None, 3)
		Total params: 63,884	

B. KEY INFLUENCING FACTORS IN INDDNN LAYERS

The performance of INDDNN layers is significantly impacted by specific parameters and processes as follows:

1) The stride parameter, a hyperparameter of the neural network's filter, governs the extent of displacement across the input data at each temporal increment. In our INDDNN model, a stride value of 1 is employed to maximize the filter's movement across the data, thereby facilitating a more comprehensive exploration of potential features.
2) The kernel value denotes the size of the filter within the deep CNN. This filter traverses the input data, systematically scanning and transforming the values into either smaller or larger magnitudes. In the proposed system, a kernel value of 3 is employed.
3) Padding, a technique employed to augment the operational domain of the neural network, facilitates increased maneuverability of the kernel across the input data. Consequently, padding is integrated into the proposed model.
4) Optimization methods work by changing the neural network's parameters and minimizing the loss function, notably the weights and learning rate. The approach proposed in this paper is an Adaptive Moment Estimation optimizer, which helps compute adaptive learning rates of all parameters at different instances. This approach improves the model's convergence properties and overall performance.

C. THE INDDNN MODEL LAYERS CONFIGURATION

Compared to traditional deep learning models that extract features within similar patterns, the INDDNN model is crafted to identify similarities among disparate features. The weights of these diverse features hence range from one extreme to the other. This guarantees first that there is no longer a large extent of the need for large datasets, second that the problem of overfitting is solved, and third that it reduces computational resources required. The next paragraph sheds more light on all 23 layers of this unique architecture:

1) Convolution 1D Layer: This layer is configured for several filters equal to 16; each has a kernel size of 3. This means each filter acts on a contiguous triplet of elements in the input and strides by a step of one because stride is set to 1. The valid padding scheme is applied, so no padding elements are added, and the

dimensions remain the same after the feature extraction process in the feature extraction layer.

2) MaxPooling layer: This layer downsamples by selecting the highest value in each pooling window. The pool size is equivalent to one, as is the stride length. More importantly, this layer's pooling does not reduce output size or create overlapping adjacent pooling windows; rather, it is a feature selection layer.
3) 1D Convolution layer: This layer convolves the output received from the prior layer by using 32 filters, each of size 3, with a "valid" padding scheme. The layer returns a feature map after convolution; hence, this is a feature extraction layer. Also used is the ReLU activation function for handling the slope for negative values, so this is a feature normalization layer.
4) The MaxPooling layer: to conduct dimensionality reduction, thereby serving as a feature selection layer.
5) The Convolution 1D layer: This layer utilizes 64 filters, each with a Kernel size of 3, a ReLU activation function, and a 'same' padding employed, to reinforce and enhance the extraction of relevant features, thereby serving as a feature extraction layer.
6) The MaxPooling layer: to conduct dimensionality reduction serving as a feature selection layer.
7) The MaxPooling layer: to conduct dimensionality reduction serving as a feature selection layer.
8) The Convolution 1D layer: This layer utilizes 32 filters, each with a Kernel size of 3, a ReLU activation function, and a 'same' padding employed, thereby serving as a feature extraction layer.
9) MaxPooling layer: for reducing the dimensionality that will act as a feature selection layer.
10) Convolution 1D layer: Uses 64 filters, each with a Kernel size of 3 and a ReLU activation function with "same" padding used, thus acting as a feature extraction layer.
11) MaxPooling layer: for reducing dimensionality that will act as a feature selection layer.
12) Convolution 1D layer: This layer has 64 filters, each with Kernel size 3, an activation function of ReLU, and "same" padding is used; thus, it acts as a feature extraction layer.
13) The MaxPooling layer: for dimensionality reduction, acting like the feature selection layer.
14) The Convolution 1D layer: This layer uses 32 filters of Kernel size 3, with ReLU as the activation function; hence, "same" padding is used, making the layer feature extracting in nature.
15) The Convolution 1D layer: This layer functions as a feature extraction layer by employing 32 filters, each with a Kernel size of 3, a ReLU activation function, and a 'same' padding.
16) The MaxPooling layer: to conduct dimensionality reduction serving as a feature selection layer.
17) Dense 1D layer: This layer receives the output from the previous layer as its input. The neurons calculate a weighted sum of inputs from the previous layer and then apply a linear activation function. During training, it acquires knowledge of the weights of connections between neurons in this layer and the neurons in the previous layer. The output size of this layer is specified by the number of units given here, which is 128. Notably, the linear activation function simply returns the input; hence, one should not overlook that this layer linearly transforms the input without any nonlinear activation. A subsequent layer may be added to include nonlinearities or other operations on the result of this layer, depending on how the model has been designed. This layer serves as a feature collection layer.
18) One-dimensional dense layer: This layer will receive the output from the previous layer as its input. Every neuron calculates the weighted sum of inputs received from the previous layer by utilizing a linear activation function during the training process. The size of the output for the current layer is determined by the number of units, which is 64 in this case. Particularly noteworthy, the linear activation function indicates that the layer operates on the input in a linear manner. Based on the structure of your model, you have the option to incorporate additional layers to introduce non-linearity or perform other operations on the output of this layer. It functions as a layer for collecting features.
19) Dense 1D layer: The output from the previous layer is taken as input by this layer. Each neuron performs a weighted summation of inputs from the previous layer and then applies the linear activation function. During training, the weights on the connection between neurons in this layer and those from the previous layer are learnt. The output size of the layer is set to 64 units. It is worth mentioning that a linear activation function indicates that it is a linear transformation of the input, without any nonlinear activation being applied to it. Additional layers can be included in the model's architecture to introduce nonlinearity or perform other operations on the output of this layer. This is similar to a layer of feature collections.
20) Dense 1D layer: This layer takes the output of the previous layer. The neurons calculate a weighted sum of the inputs from the previous layer and then apply a linear activation function. Throughout the training process, the connections between neurons in this layer and the neurons in the previous layer are learned. The output size of this layer is determined by the number of units, which in this instance is 16. This is calculated by adding the number of classes plus five, and then multiplying the result by 5. In this case, the parameter 5 represents the values for x, y, width, height, and the confidence of the model. It is important to understand that the linear activation function in this layer results in a linear transformation of the input, without any additional nonlinear activation. In the future, more layers may be added based on the model's architecture

to introduce non-linearity or perform other operations on the output of this layer. This layer functions as a collection of features.

21) The Convolution 1D layer: This layer utilizes 55 filters, each with a Kernal size of 3, a linear activation function, and a 'same' padding is employed, thereby serving as a feature extraction layer.
22) Flatten layer: This layer is responsible for transforming the input tensor into a vector with a single dimension. This is typically performed just prior to the data being inputted into a fully connected or dense layer, allowing the network to process the information in a format suitable for subsequent computations.
23) One-dimensional dense layer: This layer is added with units equal to the number of different classes and activated using the softmax method. This layer is the final step in classification tasks, and it uses the softmax function to calculate the probability distribution across the target classes. The SoftMax function ensures that the output values are always normalised; they all fall within the range of 0 to 1, and the sum equals 1. This allows the model to generate a probabilistic prediction for each class, indicating the likelihood that the input belongs to that specific class.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, the empirical validation of the INDDNN algorithm will be performed based on a set of experiments that run on three different datasets. The first dataset is based on the Bielefeld University 2020 research program [25] and contains three indoor scenarios classified as LOS, MP, and NLOS with different levels of data balance and sampling: Ds1-1, unbalanced; Ds1-2, under-sampled; and Ds1-3, up-sampled. The second open-source UWB dataset, Ds2, originates from the IEEE Data port [34] and comprises three classes: LOS, soft-NLOS, and hard-NLOS. The third one, Ds3, is a dataset collected from our experimental processes, containing four propagation channels: LOS, MP, soft-NLOS, and hard-NLOS. The last open-source dataset (Ds4) [35] was obtained from the eWINE project which measured across seven distinct indoor environments. Notably, each scenario within the dataset contains 3000 measurements for both LOS and NLOS propagation conditions.

On this front, an exhaustive set of performance metrics is used to evaluate and further validate this new multiclass NLOS identification model for achieving high accuracy. These measures include classification accuracy, precision, recall, F-measure, and computational processing time. The implemented model was run on a high-performance Lenovo laptop equipped with an Intel Core i7, 12th generation, 12700HQ processor, having a clock speed of 2.7 GHz and an 8 MB cache, besides 16 GB of RAM, with an NVIDIA GeForce RTX 3050 Ti as the graphics processor. The implementation made use of both Python and MATLAB R2022a programming languages. The model was trained for 500 epochs with an epoch size of 64, 63,000 parameter combinations, and a learning rate of 10^−3.

A. RESULTS OF THE PROPOSED INDDNN

The initial three experiments were conducted using an open-source UWB CIR dataset from Bielefeld University (2020). This dataset includes three distinct types of collected data (Ds1-1, Ds1-2, and Ds1-3), with UWB CIR data obtained using the DW1000 UWB module. Data collection occurred in three scenarios: two small rooms, a hall, and four corridors, spanning seven different indoor locations. The two rooms were a 6 m × 6 m laboratory environment and an approximately 8 m × 6 m communication room, both furnished with various items. Data collected in narrow corridors were intended to represent multipath (MP) conditions, where direct LOS was indistinguishable due to multiple signal reflections from the narrow walls. Data were collected under both static and dynamic conditions in all scenarios. The first dataset (Ds1-1) comprises 61,930 samples for LOS, MP, and NLOS conditions, resulting in a total undersampled dataset of 188,791 samples. The second dataset (Ds1-2) includes 111,468 samples for LOS, MP, and NLOS conditions, for a total of 335,539 samples. The third dataset (Ds1-3) contains 273,041 samples, including 99,264 samples for LOS, 111,846 samples for MP, and 61,930 samples for NLOS. Fig. 5 and Table 6 illustrate the classification performance of the INDDNN scheme based on Ds1-1.

TABLE 6. Classification performance of INDDNN scheme based on Ds1-1.

	Precision%	Recall%	F1-score%	Accuracy%
LOS	98.0	97.0	97.0	98.0
MP	96.0	99.0	99.0	98.0
NLOS	99.0	98.0	99.0	98.0

Based on Ds1-1, the INDDNN performs exceptionally well under LOS conditions at an accuracy of 98%. The precision is 98%, meaning that of all predictions as LOS conditions, 98% are so. This high precision means the model performs very well in avoiding false positives. It has a recall of 97% for the LOS category, which means it can capture 97% of the actual instances of LOS, showing a rather strong ability to capture true positives. The F1 score is the harmonic mean of precision and recall, 97%, thus indicating balanced and reliable performance in LOS classification. The model achieved high accuracy at 98% and F1-score of 99% in identifying MP conditions, demonstrating robust performance with a slight dip in precision to 96% due to increased false positives compared to LOS. The model consistently achieves an accuracy of 98%, which is indicative of how the model could treat all signal conditions in general. The precision for NLOS conditions stands at 99%, a value that means that the performance in predicting instances of NLOS is excellent with very few false positives. The recall for NLOS is 98%, which means that the model is very capable of correctly identifying the actual

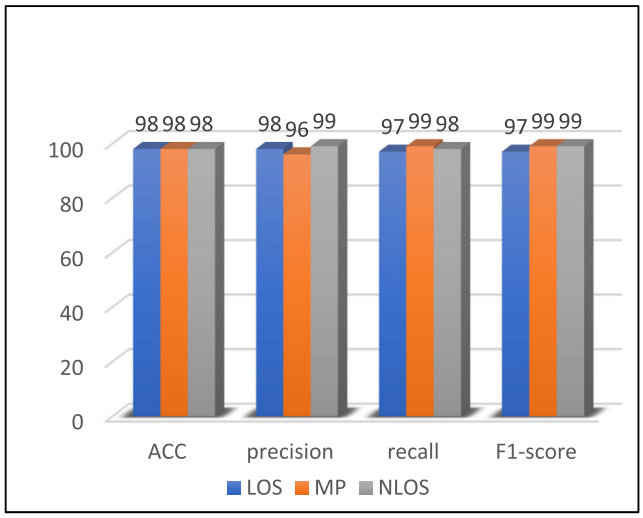


FIGURE 5. Results of the proposed model on Ds1-1.

instances of the NLOS and hence minimizing false negatives. This corresponds to an F1-score for NLOS of 99%, which means that there is a balance between precision and recall, thus proving the effectiveness of the model concerning classifying the NLOS condition. The classification performance of the proposed scheme for Ds1-2 is presented in Fig. 6 and Table 7.

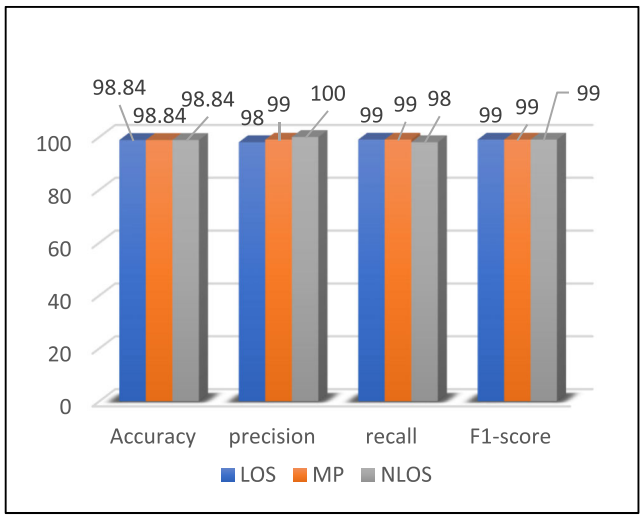


FIGURE 6. Results of the proposed model on Ds1-2.

TABLE 7. Classification performance of INDDNN scheme based on Ds1-2.

	Precision%	Recall%	F1-score%	Accuracy%
LOS	98.0	99.0	99.0	98.8
MP	99.0	99.0	99.0	98.8
NLOS	100	98.0	99.0	98.8

Across Ds1-2, the model achieved exceptional performance in LOS conditions with 98.84% accuracy, 98% precision, and 99% recall, resulting in an F1-score of 99%. This demonstrates highly effective LOS classification with minimal errors. The model maintains exceptional performance under MP conditions with 98.84% accuracy, near-perfect precision, recall (both 99%), and an F1-score of 99%, demonstrating highly reliable MP classification. The model maintains high accuracy at 98.84% and perfect precision at 100% for NLOS, with a slight decrease in recall at 98%. Overall, the model effectively classifies NLOS conditions with a near-perfect F1-score of 99%. Based on the Ds1-3, the proposed scheme's classification performance is illustrated in Table 8 and Fig. 7.

TABLE 8. Classification performance of INDDNN scheme based on Ds1-3.

	Precision%	Recall%	F1-score%	Accuracy%
LOS	99.0	96.0	97.0	98.4
MP	97.0	99.0	98.0	98.4
NLOS	99.0	99.0	99.0	98.4

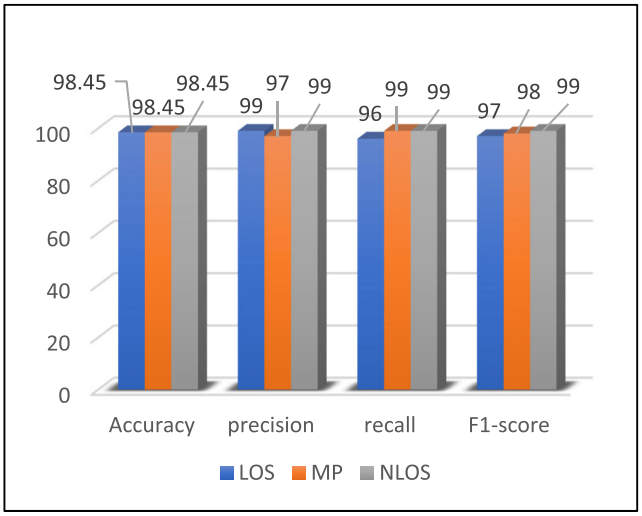


FIGURE 7. Results of the proposed model on Ds1-3.

Within Ds1-3, the model exhibited consistent performance across LOS, MP, and NLOS conditions, attaining an overall accuracy of 98.45%. Precision metrics were notably high, reaching 99% for both LOS and NLOS scenarios, and 97% for MP conditions. Recall values were similarly impressive, with 99% for MP and NLOS, and 96% for LOS. Consequently, F1-scores of 97% (LOS), 98% (MP), and 99% (NLOS) were achieved, demonstrating the model's robust and balanced classification capabilities with minimal false positives and false negatives. In addition, a separate experiment was conducted using a separate open-source UWB CIR dataset (Ds2) from the IEEE Data port, consisting of 30,732 samples across four features. This dataset includes 10,234 samples for LOS, 10,244 samples for hard-NLOS, and 10,244 samples for soft-NLOS. These samples were

collected during a measurement campaign in the scientific area building at the University of A Coruña, Spain.

The accuracy of the proposed approach using Ds2 is shown in Table 9 and Fig. 8.

TABLE 9. Classification performance of INDDNN scheme based on Ds2.

	Precision%	Recall%	F1-score%	Accuracy%
LOS	100	100	100	99.9
Hard-NLOS	100	100	100	99.9
Soft-NLOS	100	100	100	99.9

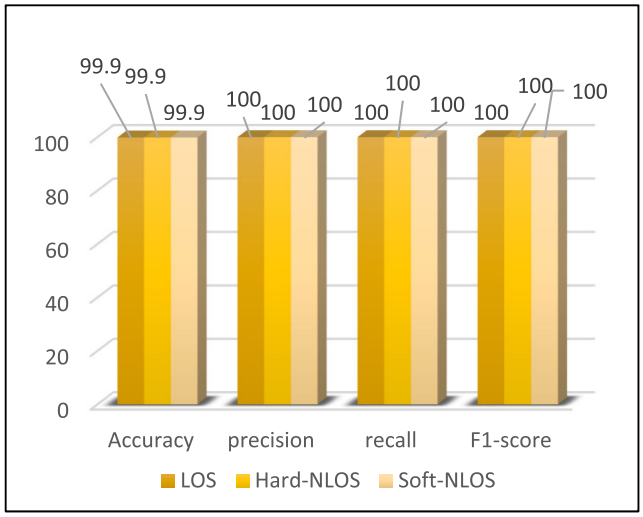


FIGURE 8. Results of the proposed model on Ds2.

On Ds2, the model performed exceptionally well under a variety of conditions. Under LOS conditions, it achieved an accuracy of 99.9%, with precision, recall, and F1-scores all at 100%, indicating flawless classification. In hard-NLOS scenarios, the model maintained a 99.9% accuracy, again with perfect precision, recall, and F1-scores, confirming its reliability in distinguishing true positive instances without false positives. Similarly, under soft-NLOS conditions, the model sustained its high accuracy of 99.9% and achieved perfect scores in precision, recall, and F1-measures. These results underscore the model's robust capability and reliability in accurately classifying signal conditions across all tested scenarios. On Ds3, the INDDNN scheme outperforms all metrics measured for LOS, MP, soft-NLOS, and hard-NLOS. The scheme's reliability and robustness are demonstrated by its consistent 99.9% accuracy across all conditions. Perfect precision, recall, and F1 scores of 100% in all categories demonstrate that the model is the best at detecting false positives and false negatives.

The proposed classification scheme is highly effective in accurately distinguishing between various signal conditions and is exemplary in its practical application, as demonstrated by these results. Table 10 and Fig. 9 demonstrate the

TABLE 10. Classification performance of INDDNN scheme based on Ds3.

	Precision%	Recall%	F1-score%	Accuracy%
LOS	100	100	100	99.9
MP	100	100	100	99.9
Soft-NLOS	100	100	100	99.9
Hard-NLOS	100	100	100	99.9

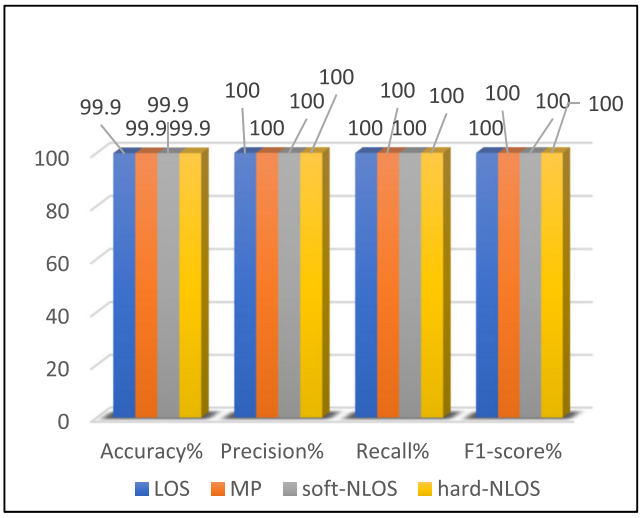


FIGURE 9. Results of the proposed model on Ds3.

identification accuracy performance of the proposed approach based Ds3.

Likewise, the INDDNN scheme on Ds4 demonstrates outstanding performance across all measured metrics for LOS, MP, soft-NLOS, and hard-NLOS. The consistent 99.9% accuracy in all conditions is a clear testament to the reliability and strength of the scheme. The model's optimal performance in identifying false positives and false negatives is demonstrated by its 100% F1 scores, recall, and precision in all categories. The results indicate that the proposed classification scheme is highly efficient in accurately distinguishing between various signal conditions and is commendable for practical applications. Table 11 and Fig. 10 demonstrate the identification accuracy performance of the proposed approach based on Ds4.

TABLE 11. Classification performance of the INDDNN scheme based on Ds4.

	Accuracy%	Precision%	Recall%	F1-score%
LOS	99.9	100	100	100
NLOS	99.9	100	100	100

B. COMPARISON OF RESULTS FOR INDDNN TECHNIQUE USING DIFFERENT DATASETS

This comparison of the INDDNN performance across the datasets Ds1 series, Ds2, Ds3, and Ds4 in the aforementioned

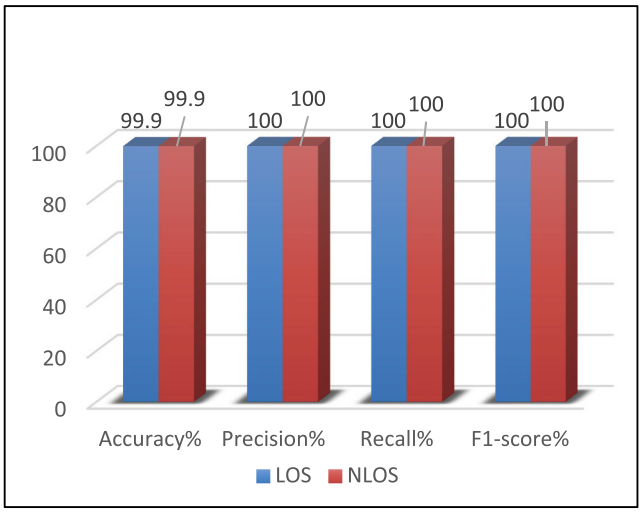


FIGURE 10. Results of the proposed model on Ds3.

tables and figures shows excellent performance with faintly varying trends in accuracy, precision, recall, and the F1-score, besides a difference in the efficiency of training. Among all other Ds1 series, Ds1–2 has the best performance, as it recorded the highest average accuracy of 98.84% and an F1-score of 99%, thus representing improved precision and recall compared to Ds1-1 and Ds1-3, which had slightly lower metrics. While the proposed model exhibited perfect distinctive performance across all the datasets, Ds2, Ds3, and Ds4 perform better compared with the Ds1 series, achieving perfect scores in precision, recall, and F1-score with an average accuracy of 99%, also coupled with perfect precision, recall, and an F1 score of 100%, depicting its outstanding performance. It also displays the high training efficiency of Ds3, and Ds2, with only 1 second per epoch for training, far ahead of the rest. while the implementation of the INDDNN takes 16s across Ds4. Based on these results, the INDDNN model presents a superior accuracy in identifying LOS/NLOS conditions. Notably, the model prioritizes low computational complexity in both its temporal and structural design. Furthermore, the model demonstrates strong generalization capabilities, performing effectively across various datasets and environmental configurations, making them more appropriate for real applications that require high precision and faster processing. Table 12 presents a comparison of proposed model performance across various datasets.

C. COMPARISON WITH RELATED STUDIES

Table 13 compares the accuracy achieved by different methods. Such a comparison clearly demonstrates the diversity of techniques and performance in terms of accuracy, precision, recall, F1-score, and total time in seconds available for various studies. The INDDNN technique, applied differentially across datasets Ds1, Ds2, Ds3, and Ds4, outperforms others in terms of accuracy, precision, recall, and F1 score. In particular, achieves nearly perfect results with an accuracy

TABLE 12. Comparison of results for INDDNN technique using different datasets.

	Avg. Accuracy	Avg. Precision	Avg. Recall	Avg. F1-score%	Loss Function	No. Epoch	Time/ epoch
Ds1-1	98.0	97.6	98.0	98.3	0.030	500	6s
Ds1-2	98.8	99.0	98.6	99.0	0.039	500	8s
Ds1-3	98.4	98.3	98.0	98.0	0.040	500	10s
Ds2	99.0	100	100	100	0.003	500	1 s
Ds3	99.9	100	100	100	0.002	500	1 s
Ds4	99.9	100	100	100	0.000	500	16s

of 99.98%, perfect precision, recall, and F1-scores of 100%, and an impressively short training time of only 1 second per epoch. It outperforms other techniques, such as MIPL-B (96.05% accuracy), ST, CNN, and TL (97% accuracy), and ODWPA, CNN (variable scene accuracy). However, in a different scenario, the Bo-FDT showed a decrease in NLOS identification accuracy, reaching 89.5%. Similarly, CNN-LSTM detects 95% of NLOS in the investigated environment. Other techniques, such as SVM, GA (91.86%), and S3VM (94.72%) for NLOS, perform poorly when compared to this high performer INDDNN on various investigated datasets. The total time metrics also mean that the INDDNN is not only effective but very efficient in contrast to the longer times usually reported by techniques like ININN at 57.13 seconds. Overall, the INDDNN method gives outstanding classification performance and efficiency, hence very suitable for real-world applications where both accuracy and speed are important. As detailed in Appendix A, the INDDNN model's performance significantly surpasses that of other state-of-the-art networks. Table 13 in Appendix A provides a comprehensive comparison of the model's accuracy, precision, recall, F1-scores, and computational complexity against established models like CNN-LSTM, ODWPA-CNN, SVM-GA, and others. The results consistently demonstrate the INDDNN's superiority, achieving near-perfect accuracy across multiple datasets while maintaining lower computational costs. This validation underscores the model's robustness and generalizability, establishing a new benchmark for NLOS identification in UWB positioning systems.

VII. CONCLUSION

In this study, the quality of the method is guaranteed by the INDDNN classification scheme, which identifies the signals correctly across various datasets. The method showcases exceptional performance in terms of accuracy, precision, recall, and F1-score when compared to existing techniques. The INDDNN model applied particularly to the different datasets delivered close-to-perfect results. This will be more clearly realized on the dataset collected from our experimental processes, which showed an accuracy of 99.98% with perfect precision, recall, and F1-scores of 100%, not forgetting an impressively low training time of just 1 second per epoch. These results underline the ability and reliability

TABLE 13. Comparison of recent results for different techniques using UWB technology.

Ref. NO.	Techniques	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Total Time (s)
[13]	MIPL-A (GAF, CNN, ResNet) MIPL-B (GAF, CNN, ResNet)	89.5 96.05	90.26 96.28	90.16 96.19	90.18 96.24	… …
[14]	ST, CNN, TL	97	97.51	97.07	97.10	29-58
[15]	ODWPA, CNN	Scene#1 = 100 scene#2 = 95 scene#3 = 90	scene#1=100 scene#2=98.1 scene#3=92.5	scene#1=100 scene#2=98.1 scene#3=92.3	scene#1=100 scene#2=95 scene#3=90	38.793
[16]	CNN, LSTM, fully connected layers	LOS = 99 NLOS = 96	…	…	LOS= 99 NLOS= 95	…
[18]	BO-FDT	Scene #1 LOS=99 NLOS=98.6 Scene #2 LOS = 96.83 NLOS =89.50	…	…	…	…
[19]	DTC, FC-SVM	93.27	…	LOS=94.72 NLOS=92.57	…	…
[20]	ININN (CNN, GRU, SE)	#Open-source dataset = 88.45 #at six different scenes =87.5 to 99.9	…	…	…	57.13
[21]	MIL, CIR, TFDOCIR	91.74	…	…	…	…
[22]	GD, and GGD	98	…	…	…	…
[23]	MWT-CNN	scene#1= 98.84 scene#2= 76.65 scene#3= 69.23	…	…	…	…
.[24]	S3VM	NLOS=94.72 LOS=77.2	…	…	…	…
[25]	SVM RF MLP	82.8 91.9 91.2	…	…	83 92 91	…
[26]	CNN	98.24	…	…	…	…
[27]	CapsNets	94.63	LOS=93.48 NLOS=95.58	LOS=94.51 NLOS= 94.74	94.90	..
[28]	SVM, GA	91.86	…	…	…	…
[29]	FCN-Attention	#Local dataset = 100 #eWINE dataset = 88.24 #Mixed data = 92.01	--	#Local dataset =100 #eWINE dataset = 91.56 #Mixed data = 94.45	#Local dataset =100 #eWINE dataset =88.62 #Mixed data = 92.15	…
Ds1-1	INDDNN	98	97.66	98	98.33	6
Ds1-2	INDDNN	98.84	99	98.66	99	8
Ds1-3	INDDNN	98.45	98.33	98	98	10
Ds2	INDDNN	99	100	100	100	1
Ds3	INDDNN	99.98	100	100	100	1
DS4	INDDNN	99.9	100	100	100	16

TABLE 14. List of abbreviations.

Abbreviation	Definition
NLOS	Non-Line-of-Sight
UWB	Ultra-Wide Band
LOS	Line-of-Sight
CIR	Channel Impulse Response
DNN	Deep Neural Network
IPs	Indoor Positioning System
IoT	Internet of Things
INDDNN	Indoor NLOS/LOS Detection Deep Neural Network
GAF	Gramian Angular Field
ST	Stockwell Transform
ODWPA	One-Dimensional Wavelet Packet Analysis
LSTM	Long Short-Term Memory
CCP	Characteristic Parameters
ININN	Indoor NLOS/LOS Identification Neural Network
GRU	Gated Recurrent Unit
SE	Squeeze-and Excitation
Eb/N0	Energy per Bit to Noise Power Spectral Density Ratio
TFDOCIR	Time-Frequency Diagram of CIR
CWT	Continuous Wavelet Ttransforms
GD	Gaussian Distribution
GGD	Generalised Gaussian Distribution
MWT	Morlet Wavelet Transformer
S3VM	Semi-Supervised Support Vector Machine Classifier
SVM	Support Vector Machine
RF	Random Forest
MLP	Multilayer Perceptron
CapsNets	Capsule Networks
STFT	Short-Time Fourier Transform
GA	Genetic Algorithm
FCN	Fully Convolution Network
TW-TOA	Two-Way Ranging-Time of Arrival
MCU	Microcontroller
UART	Universal Asynchronous Receiver-Transmitter
USB	Universal Serial Bus
USART	Universal Synchronous/Asynchronous Receiver/Transmitter
PC	Personal Computer
Conv1D	Convolution One Dimension
DS	Dataset
CNN	Convolutional Neural Network
TL	Transfer Learning

of INDDNN against a changing signal environment including Line-of-Sight, Non-Line-of-Sight, and Multipath conditions. Comparative analysis with other state-of-the-art methods like MIPL-A, (ST, CNN, TL), (ODWPA, CNN), and (CNNLSTM) shows that INDDNN performs better. Other approaches may have decent accuracy and precision, but they often lack the same level of balance, generalization, and efficiency as INDDNN. In addition, the INDDNN demonstrates exceptional performance across various datasets, showcasing its adaptability and potential for practical use. Ultimately, the INDDNN classification scheme has developed a highly effective signal classification method that significantly enhances the performance of current techniques. With high-performance metrics and training efficiency, this can be a potential tool for applications in communication and navigation systems that require prompt and accurate signal classification. In the future, using this approach, its scalability could be studied and integrated into an even more complex and diversified real-world scenario with enhanced power and strength.

Future work will benefit from investigating the INDDNN model's scalability on larger and more complex datasets, examining the model in applications such as real-world communication systems and autonomous vehicles, and assessing its adaptability across new and variable signal environments. Further, studying the optimization techniques for training and inference times, comparison of the model with emerging machine learning methods, and its interpretability to enhance robustness to adversarial attacks, would add more value to the findings. Besides, it can offer major improvements and practical benefits by generalizing the model's applicability to multi-modal learning frameworks and user-centered evaluations.

APPENDIX

See Tables 13 and Table 14.

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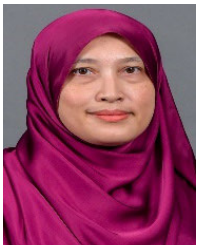
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